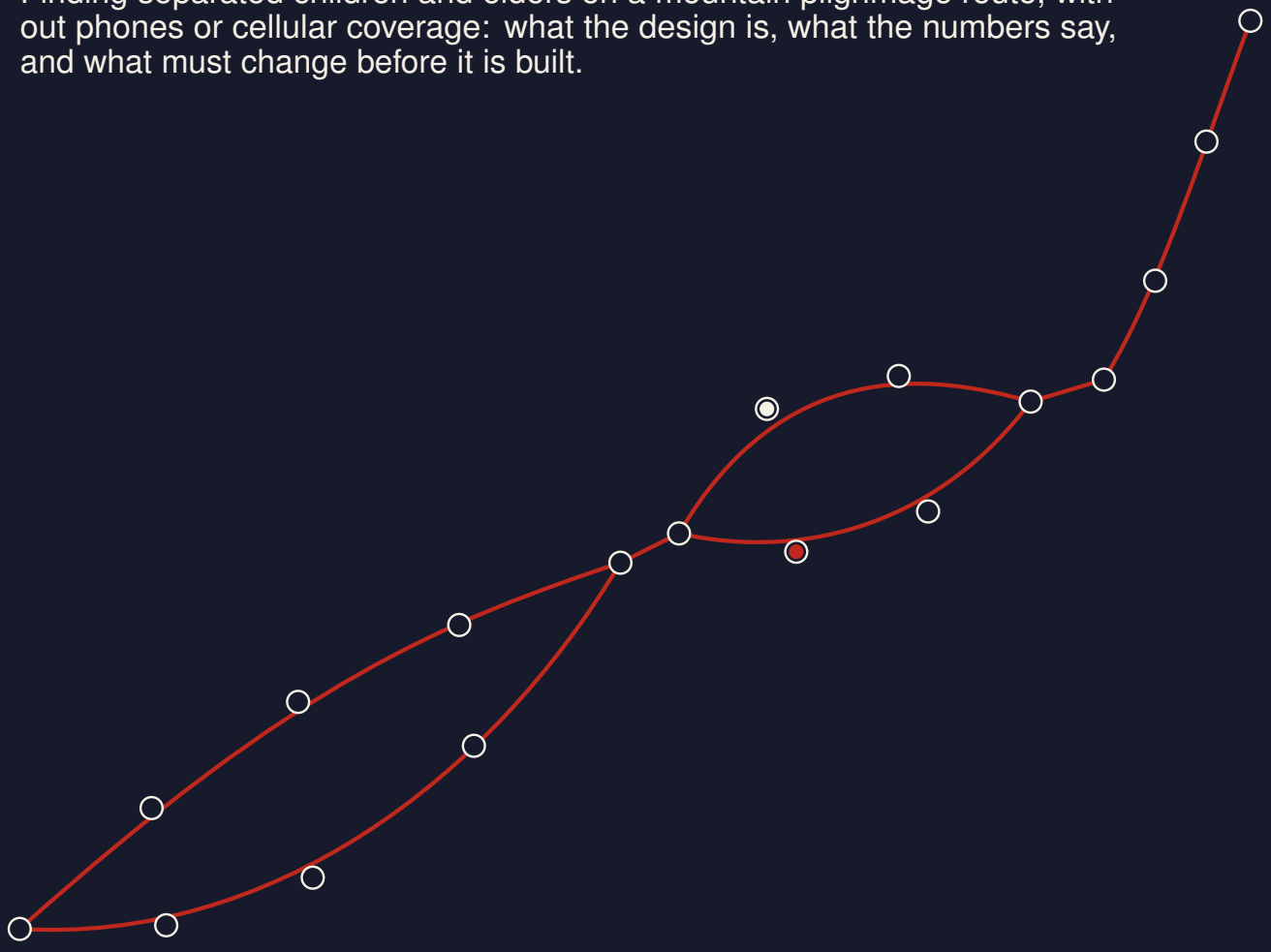
RAKSHASUTRA DESIGN, MODEL AND AUDIT

A Thread Through the Crowd

Finding separated children and elders on a mountain pilgrimage route, without phones or cellular coverage: what the design is, what the numbers say, and what must change before it is built.



Raj Jha

Independent researcher | First edition, November 2026

Every result in this book is calculated or simulated. Nothing has been built or measured. Not peer reviewed.

A Thread Through the Crowd. Rakshasutra: design, model and audit.

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Code, data, results and the source of this book: https://github.com/0xr4j/rakshasutra
Website: https://0xr4j.github.io/rakshasutra/

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This book describes a design. No hardware has been built and nothing has been measured on any route. It is not a product specification and not safety advice, and it has not been peer reviewed. Place names follow the Shri Mata Vaishno Devi route, which serves as the model; the work was not commissioned or endorsed by the Shrine Board.

Preface

A raksha sutra *is the thread tied on a wrist for protection. This book is about an electronic one.*

The idea came from a simple scene. A family walks a crowded mountain path to a shrine. A child lets go of a hand. Phones are in a locker at the top of the hill, the network is down in the valley, and the path forks twice. The cards issued at the gate will say that the child entered, and nothing more.

I wrote a white paper describing a system for that scene: a wristband that announces itself by Bluetooth, poles along the path that listen, and long-range radio to carry what they hear to a control room. The paper was full of numbers, and when I set out to publish it properly, with code that anyone could run, I found that I could not reproduce all of them. Some were right to the last digit. Some were wrong. Three mechanisms did not work as I had described them.

So this book is two things. It is a full account of the design, written so that a student of electronics or computer science can follow every step. And it is an audit: every checkable number in the first draft of the paper, recomputed by open code, with the verdict stated. Where the first draft was wrong, this book says so and uses the corrected value. I would rather publish a design with its faults marked than one that only looks finished.

The study in numbers

15	bytes in the signed advertisement a wristband sends every 800 ms
255	poles on the 22.5 km model route, 195 on LoRaWAN and 60 on fibre
57.6%	of uplink frames lost in a 3,000-wearable surge if every arrival is its own uplink [calculated]
21.3%	lost with 25-tag chunks and a 64-byte Bloom heartbeat, without departure lists [calculated]
0.5 s	median time for an SOS to reach the control room; 92% arrive within 4 s under a surge [simulated]
12.8	months of battery on a coin cell at 25 °C on Coded PHY; 9.0 at -5 °C [calculated]
42	claims of the first draft audited: 16 reproduced, 16 differ, 6 need a design fix, 4 open
104	automated tests, including a byte-for-byte comparison of the C firmware core with the Python reference
0	measurements. Everything here is design, calculation or simulation

Five things to take away

1. A wristband that only announces itself, heard by poles that already stand along the shelters, can follow a person between gates without a phone, a SIM or a satellite fix.
2. Privacy and authentication fit in 15 bytes: an identifier and a radio address that change every 15 minutes, and a truncated AES-CMAC that lets the server reject a forged alarm.
3. What limits the system in a surge is LoRa airtime. Batching arrivals does most of the work; a fixed-size Bloom filter keeps every frame legal; departure lists cost more than they are worth.
4. Three parts of the first design do not work as specified: the tether's 5 ms scan, the 100 μF capacitor meant to carry the buzzer in the cold, and unsigned sightings, which can be replayed.
5. None of this has met a real steel roof. The first measurement to make is the simplest: how far a wristband carries under a shelter, and how much signal the roof takes.

Notation

Symbol	Meaning
$G = (V, E, W)$	the route: poles, edges between adjacent poles, edge lengths in metres
$d_G(a, b)$	shortest distance along the track between poles a and b
K_{master}, K_u	master key held by the server; key of wearable u
c	16-bit slot counter of a wearable; increases every 900 s
$\text{ID}_{u,c}$	32-bit rotating identifier (Rolling_ID) of wearable u in slot c
MIC	message integrity code: the first 32 bits of an AES-CMAC
$\text{CMAC}_K(m)$	AES-CMAC of message m under key K (RFC 4493)
P_{tx}, P_{rx}	transmitted and received power, dBm
L_{GI}, L_{body}	loss through the steel roof; loss through bodies, dB
n, σ	path-loss exponent; standard deviation of shadowing, dB
SF, BW	LoRa spreading factor and bandwidth
$T(B)$	time on air of an uplink with B bytes of application payload
G (load)	offered load: airtime per second, per channel
N, K, τ	wearables in a cell; poles in a cell; mean time between poles
H	decoded frames a pole needs before it reports an arrival

Results carry one of two labels. [calculated] means the number follows from a formula in this book. [simulated] means it comes from a Monte Carlo run with a fixed seed. There is no third label, because nothing has been measured.

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CHAPTER 1

The problem

Four ordinary facts about a mountain shrine defeat every tracker you can buy.

1.1 A crowded path

The Shri Mata Vaishno Devi shrine stands in the Trikuta hills of Jammu and Kashmir. About 9.5 to 10 million pilgrims visit in a year. From the base at Katra the path climbs from about 762 m to the shrine complex, the Bhawan, and beyond it to Bhairon Ghati at 2,017 m. There are two parallel tracks for the first half, a junction at Adhkuwari, two tracks again, then the Bhawan. One of the upper tracks carries electric vehicles. In all there are about 22.5 km of path.

In such a crowd children and old people get separated from their groups, and some need medical help far from a checkpoint. The shrine already has registration counters, cards, cameras and announcements. What it lacks is a way to answer one question quickly: *where on the path is this person now?*

1.2 Four constraints

No usable phone network. Cellular networks on the route are congested during festivals and shadowed by ridges. A tracker that reports over a SIM cannot be relied on. Consumer Bluetooth tags do not report at all by themselves: they are located when a stranger's phone hears them and uploads the sighting, which needs both the phone and its data connection.

Passive cards see only the gates. The registration card, the *Yatra Parchi*, is a passive RFID tag. It has no battery and answers a reader within 1 to 3 m. It records that a pilgrim passed a gate. Between two gates, kilometres apart, it records nothing.

Steel shelters. To protect pilgrims from rain and rockfall, long stretches of the path are covered by galvanised iron sheds. A grounded steel roof is a good radio shield. A satellite receiver beneath it sees little sky, and a radio above it hears little from below.

No phones near the sanctum. Phones and bags are left in cloakrooms before the inner complex. So even a found child with a phone number printed on a wristband cannot be reunited by a call: the parent's phone is in a locker.

1.3 What exists, and where it stops

The fifth column deserves a word. Combining Bluetooth beacons with LoRaWAN is not new. The usual scheme forwards each beacon a relay hears, with its address and signal strength, as a LoRaWAN uplink. With a few dozen beacons this works. With thousands it fails in three ways at once, and each failure shapes a chapter of this book: the beacon's fixed address lets anyone follow it (Chapter 3); the uplinks collide and overflow the payload limit (Chapters 5 and 6); and the reply path is too slow to reach someone who is walking (Chapter 7).

Table 1.1 Approaches to finding a person on a covered mountain route. Rakshasutra's entries are design targets and calculated values.

	Passive RFID or QR	GPS and cellular	Consumer BLE tag	BLE, raw LoRaWAN	Rakshasutra
Between gates	No	Open sky only	Needs phones	Low density only	Yes, nearest pole
Without cellular	At gates	No	No	Yes	Yes
Under steel sheds	At gates	No	Poor	Poor	Split antennas
Phone-free zones	No alert	–	No	No alert	Kiosk, parent tag
Privacy	Static ID	SIM	Rotating key	None	Rotating pseudonym
3,000 in one cell	–	–	–	57.6% loss	21.3–30.5% loss
Battery	None	1–2 days	About a year	Months	9.0–12.8 months

1.4 Questions this book answers

1. Can a wristband be anonymous to a stranger and still be identified, and its alarms trusted, by the server?
2. How far does a wristband carry under a steel roof, and how should a pole be mounted?
3. How many wristbands can one LoRaWAN gateway cell carry before reports are lost, and which filtering helps?
4. Can an alert reach a person who is walking away, and can a passenger in a vehicle be seen at all?
5. How long does the wristband's battery last, and does it survive the cold?
6. Which numbers of the first white paper stand, and which do not?

CHAPTER 2

The design at a glance

A band that speaks, poles that listen, a server that knows the map.

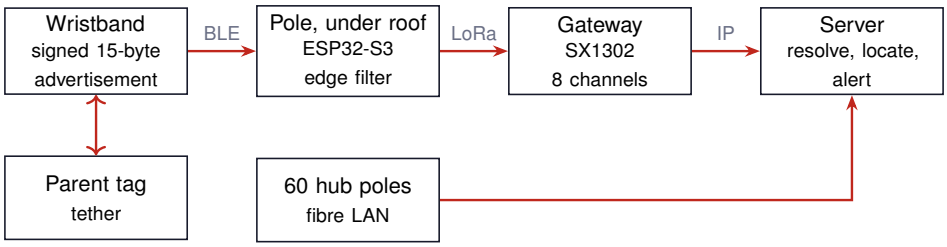


Figure 2.1 The data path. A pole listens below the roof and transmits above it. The 60 poles of the two crowded hubs bypass radio and use the shrine's fibre network.

The wristband is a coin-cell Bluetooth beacon in a sealed strap, issued at registration to a child or an elder and returned at the exit. It has a button for SOS, a buzzer and a light. It does not know where it is. It sends the same short message every 800 ms: who I am in this quarter of an hour, how my battery is, whether my button is pressed, and a signature.

The parent tag is the same hardware with a different role. It carries a list of the child band's next identifiers and buzzes when it stops hearing them.

The pole is a small box clamped under the shelter roof on an existing light pole: an ESP32-S3 that scans for wristbands, and an SX1262 LoRa radio whose antenna is led through the roof. It keeps a table of the wristbands in range and decides what is worth sending. It holds no secret that identifies anyone.

The gateways, six of them on hilltops, receive the poles' uplinks on eight channels and pass them to an on-premise server. **The server** holds the keys, the pairing records and the map of the route. It turns an anonymous identifier heard at pole 175 into "the child registered with family 0421 is on the Sanjichhat track, 243 m from her parent", and raises alarms.

2.1 The route in the model

The 195 LoRaWAN poles cover 21.0 km, one per 108 m on average. The first draft said poles stand "every 25 to 100 m"; with 255 poles in all, 25 m in the hubs and about 108 m on the tracks is what fits. Chapter 4 shows that the radio range allows it.

Table 2.1 Zones of the model route. Lengths and the pole total are the white paper's; positions within a zone are schematic.

Zone	Segments	km	Poles	Spacing (m)	Backhaul
1	old-track	6.0	56	107	LoRaWAN
2	tarakote	7.0	65	108	LoRaWAN
3	adhkuwari	0.5	20	25	fibre LAN
4	himkoti + sanjichhat	6.5	60	108	LoRaWAN
5	bhawan	1.0	40	25	fibre LAN
6	bhairon	1.5	14	107	LoRaWAN
	Total	22.5	255		

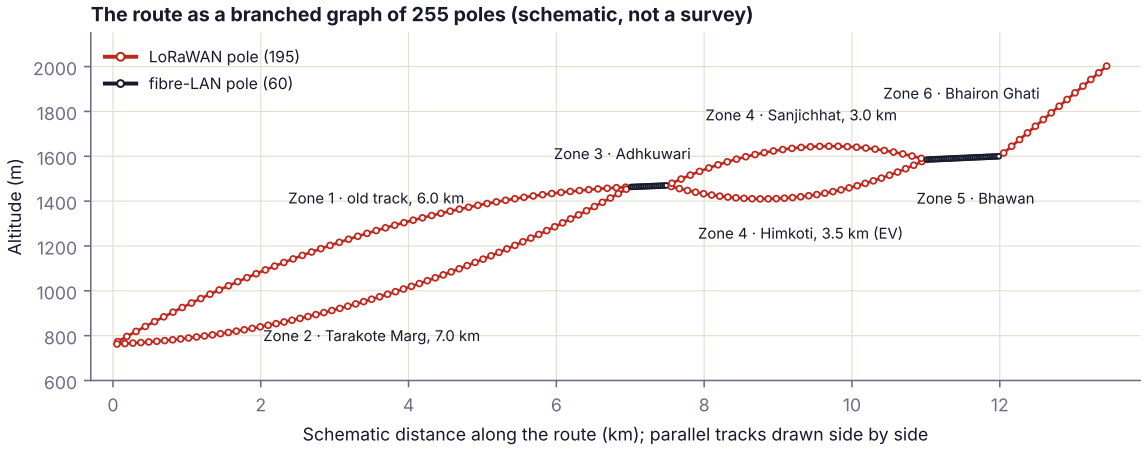


Figure 2.2 The model route. Zone 4 is two parallel tracks; their split of 6.5 km into 3.5 and 3.0 km is assumed.

CHAPTER 3

The signed advertisement

Fifteen bytes that say who is here to the server, and nothing to anyone else.

3.1 Who might attack a child's wristband

A system that tracks children must be designed against people who would misuse it. Four are considered.

- $\mathcal{A}_1$, the **stalker**, places cheap receivers along the path and tries to follow one wristband, or to learn which wristbands travel together.
- $\mathcal{A}_2$, the **replayer**, records advertisements and broadcasts them again somewhere else or later, to create false sightings or false alarms.
- $\mathcal{A}_3$, the **forger**, sends commands to a wristband to make it buzz or to flatten its battery.
- $\mathcal{A}_4$, the **thief**, takes a pole or a wristband apart to read keys from its memory.

3.2 Keys

The server keeps one 128-bit master key, K_{master}, in an encrypted keystore or a hardware security module. Each wearable gets its own key when it is made,

$$K_u = \mathrm{CMAC}_{K_{\mathrm{master}}}(\mathrm{UID}_u \,\|\, \mathrm{Salt})\,, \tag{3.1}$$

and its debug port is locked. If $\mathcal{A}_4$ extracts K_u from one wristband, that wristband is compromised and no other. A pole never receives K_{master} or any K_u, so a stolen pole reveals no identities. The pole has its own LoRaWAN keys, and those are worth protecting too, with secure boot and encrypted flash, because they let it speak to the network.

3.3 A counter instead of a clock

The wearable has no clock set to the time of day, and setting one would mean a two-way Bluetooth connection at the registration desk, which is slow in a queue. It keeps a 16-bit counter c instead, stepped every 900 s by its 32.768 kHz crystal, and *sends the counter in every frame*. The registration pad only has to listen once.

3.4 The frame

In slot c the wearable computes an identifier and a radio address,

$$\mathrm{ID}_{u,c} = \mathrm{Trunc}_{32}\,\mathrm{CMAC}_{K_u}(\texttt{0x01} \,\|\, c)\,, \tag{3.2}$$

$$M_{u,c} = \mathrm{Trunc}_{46}\,\mathrm{CMAC}_{K_u}(\texttt{0x02} \,\|\, c) \text{ with both top bits set,} \tag{3.3}$$

and broadcasts the 15 bytes of Table 3.1 as manufacturer data.

Table 3.1 The 15-byte payload.

Bytes	Field	Content
0–1	Company_ID	`0xFF5A`, a laboratory value; sent low byte first
2	V_type	bits 0–3: version 4. Bit 4: 0 for a band, 1 for a parent tag
3–4	Slot_Counter	c, little-endian
5–8	Rolling_ID	$\mathrm{ID}_{u,c}$
9	F_stat	bit 0 SOS pressed; bit 1 tether lost; bit 2 buzzer on; bit 3 low battery
10	B_mv	supply voltage in steps of 20 mV (150 means 3.00 V)
11–14	MIC	$\mathrm{Trunc}_{32}\,\mathrm{CMAC}_{K_u}$(bytes $2 \ldots 10$)

A worked example

Take the tag key `00 01 02 ... 0f`, slot 17, SOS pressed, battery 3.00 V. The reference implementation gives

```
5a ff 04 11 00 3c 34 fd c9 01 96 8c 83 e4 2f
```

and the radio address `d4:1b:8a:54:a8:dc`. The command `rakshasutra frame -key 000102...0f -counter 17 -sos` prints this, and the C firmware core produces the same bytes: a test compares the two on 200 random keys.

Three encodings the first draft left open

The paper wrote Trunc_{32} and "`0x01` $\|$ c" without saying which bytes. An implementation must choose, and two implementations must choose alike. The reference code fixes: truncation keeps the *leading* bytes of the CMAC, as RFC 4493 does; the counter enters each CMAC as the same two little-endian bytes it has on air; and the address is the first six CMAC bytes with the two most significant bits set, which is what marks a static random address in Bluetooth. These are listed as DEFINED in the parameter table of Appendix A.

3.5 How the server knows who it is

At registration the pad hears one frame, checks its MIC, and stores the counter it saw, c_{init}, and the time, t_{pair}. Afterwards the server expects, at time t,

$$\hat{c}_u(t) = \left(c_{init} + \left\lfloor \frac{t - t_{pair}}{900} \right\rfloor \right) \bmod 65536. \tag{3.4}$$

A crystal accurate to 20 parts per million drifts 1.7 s a day, so the wearable may step its counter a little before or after the server expects. The server therefore keeps three identifiers per wearable, for slots $\hat{c}_u - 1$, $\hat{c}_u$ and $\hat{c}_u + 1$, in a hash map from identifier to wearable. Resolving a report is one lookup.

> **Identifiers can clash.** The first draft treated the map as one-to-one. With 10,000 wearables and three slots each there are 30,000 entries drawn from 2^{32} values, and by the birthday bound the chance that some two coincide at a given moment is $1 - \exp(-n(n-1)/2^{33}) \approx 10\%$. The reference server keeps a list per identifier and uses the counter in the report to tell clashing entries apart. [calculated]

3.6 What an eavesdropper learns

AES-CMAC is a pseudorandom function: without the key, its outputs for different inputs look like independent random strings. So to $\mathcal{A}_1$ the identifier and address of one slot have no visible connection to those of the next. A wristband can be followed for at most the remainder of one 15-minute slot, about 1.1 km of walking. Two things soften even that. The battery byte and the counter are sent in clear, and a counter that happens to be unusual could help link slots; a later revision should mask the counter. And all the wristbands of a family rotate at different instants, which hides who travels with whom from a casual listener, though not from one who watches rotations closely.

3.7 What a forger can and cannot do

The MIC covers every byte that matters, so $\mathcal{A}_2$ and $\mathcal{A}_3$ cannot alter a frame: a test flips each of the 104 bits of bytes 2 to 14 in turn and checks that verification fails every time. Guessing a 32-bit MIC succeeds once in 2^{32} tries, and a pole forwards at most ten SOS frames a minute, so a brute-force attack on one pole would need about 800 years on average.

> **A limit of the design: ordinary sightings are not authenticated.** To save airtime a pole reports an arrival as 8 bytes: identifier, counter, signal strength, flags. The MIC is left behind, and the pole, holding no key, could not have checked it anyway. So a frame recorded near one pole and replayed near another, inside its three-slot window of up to 45 minutes, is believed. It cannot raise an alarm, because an SOS travels as the whole frame and is verified. But it can put a child's marker in the wrong place. The reference server flags any sighting that implies an impossible speed, which catches a careless replay and not a careful one. The remedy is to carry the 4-byte MIC in each arrival tuple and verify it at the server; the cost in airtime is worked out in Chapter 6. The repository's example `attack_log.csv` shows all of this: forged alarms dropped, stale frames dropped, and one same-slot replay accepted and flagged.

CHAPTER 4

Radio under a steel roof

The roof that keeps the rain off also keeps the signal in. Use both sides of it.

4.1 The model

Received power falls with distance. The standard description is a straight line on a logarithmic scale:

$$P_{rx}(d) = P_{tx} + G_{tx} + G_{rx} - PL(d_0) - 10\,n \log_{10}(d/d_0) - L_{body} - X_\sigma. \tag{4.1}$$

$PL(d_0)$ is the loss at a reference distance $d_0 = 1$ m. The exponent n is 2 in free space; a corridor with a reflecting roof can guide the signal and keep n near 2 even among obstacles. L_{body} is what a crowd absorbs, and X_σ is a random term, normally distributed in decibels with standard deviation σ, for everything the line does not capture. The white paper's values are $P_{tx} = +8$ dBm, $PL(1\,\text{m}) = 40.2$ dB, $n = 2.15$, $L_{body} = 14$ dB and $\sigma = 4.2$ dB. **None has been measured under a shelter.** They are plausible design values, and everything in this chapter is conditional on them.

4.2 From margin to range

A frame is received if P_{rx} exceeds the receiver's sensitivity S. The *margin* is the median received power minus S. With the random term, a frame at margin m is received with probability $\Phi(m/\sigma)$, where Φ is the normal distribution function. For 98% delivery the margin must be $2.05\sigma = 8.6$ dB. Solving Eq. 4.1 for the distance at which that margin is left gives the reliable range.

For Coded PHY at $S = -103$ dBm and 50 m:

$$P_{rx} = 8 - 40.2 - 21.5 \log_{10} 50 - 14 = -82.7\ \text{dBm}, \qquad m = 20.3\ \text{dB}.$$

The first draft stated 28.6 dB. The difference, 8.3 dB, would need antenna gain that the paper never specifies, and a wristband antenna is more likely to lose a few decibels than to gain eight. This book uses 20.3 dB.

The ranges, 82 m on 1M PHY and 174 m on Coded PHY, are longer than the 28.5 m and 92 m the first draft tabled; its table does not follow from its own equation. The conclusion survives either way: Coded PHY covers poles 108 m apart, and 1M PHY would not with any comfort.

4.3 Why the pole is split

Mount the whole pole above the roof, and the Bluetooth signal from a wrist below must pass through steel: 24.1 dB lost by the design value. With a $+4$ dBm, 1M PHY wristband the reliable range collapses to 4.1 m. Mount the whole pole below the roof, and the LoRa signal must pass through steel on its way to the hilltop: 18.4 dB lost, and the maximum allowable path loss falls from 145.4 dB to 124.9 dB.

Table 4.1 Link budget from Eq. 4.1 [calculated]. Bluetooth rows: the distance at which 98% of frames clear the sensitivity. LoRa rows: margin at 8 km behind one assumed 20 dB ridge.

Link	PHY	P_{tx}	Roof	S_{rx}	Range or margin
Wearable to pole, pole above the roof	BLE 1M	+4	24.1	-96.0	4.1 m
Wearable to pole, pole under the roof	BLE 1M	+8	0.0	-96.0	82.2 m
Wearable to pole, pole under the roof	BLE Coded S=8	+8	0.0	-103.0	174.1 m
Pole to gateway, antenna under the roof	LoRa SF7	+14	18.4	-126.5	-4.4 dB
Pole to gateway, whip above the roof	LoRa SF7	+14	0.0	-126.5	+16.1 dB

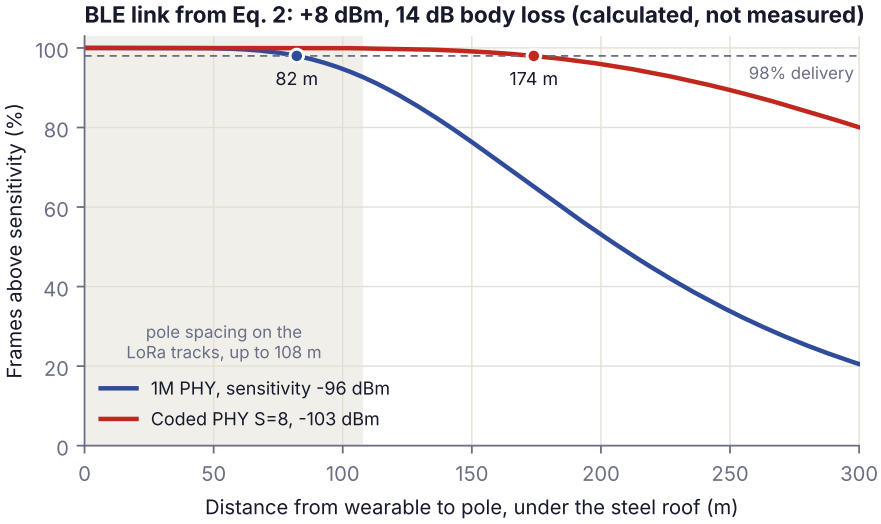


Figure 4.1 Share of frames above sensitivity against distance.

The split-antenna pole avoids both. The electronics and the Bluetooth antenna sit under the roof truss, looking down at the crowd. One metre of thin coaxial cable, costing 0.9 dB, takes the LoRa signal through a sealed gland to a whip antenna above the ridge.

The maximum allowable path loss follows directly:

$$\mathrm{MAPL} = 14 - 0.9 + 5.8 + 126.5 = 145.4 \text{ dB}.$$

How far that reaches depends on the valley, and the white paper gave ranges of 5.4 to 8.8 km without a model to derive them from. This book claims no range. As an indication only: free-space loss at 866 MHz over 8 km is 109.3 dB; add one ridge in the way at an assumed 20 dB, and the split antenna keeps 16.1 dB in hand at SF7, while the antenna under the roof is 4.4 dB short. [calculated, with an assumed ridge]

4.4 The gate that defines “here”

A pole hears wristbands far beyond its own stretch of path. To report a wristband as *here* it requires a signal above a gate: -75 dBm on the tracks, -65 dBm in the hubs. By Eq. 4.1 these correspond to about 22 m and 7.5 m. That is the meaning of the “25 m core zone” used in Chapter 7.

What to measure first. Three numbers carry this whole chapter: the roof loss at 2.44 GHz and at 866 MHz, and the exponent n under the roof. All can be measured in an afternoon with two development boards and a tape measure under any GI shed. Until they are, treat every range here as a hypothesis.

CHAPTER 5

Airtime

Every report costs milliseconds on a shared channel. Count them.

5.1 How long a LoRa frame lasts

LoRa trades speed for range. At spreading factor SF and bandwidth BW one symbol lasts $T_{sym} = 2^{SF}/BW$: 1.024 ms at SF7 and 125 kHz. A frame is a preamble, a header and the payload, coded so that four bits travel as five at coding rate 4/5. For a physical payload of PL bytes with an explicit header and a CRC, the number of payload symbols is

$$n_{pay} = 8 + \max\Big(\Big\lceil \frac{8\,PL - 4\,SF + 44}{4\,SF} \Big\rceil \cdot 5,\ 0\Big), \tag{5.1}$$

and the whole frame lasts $T = T_{sym}\,(n_{pre} + 4.25 + n_{pay})$ with a preamble of $n_{pre} = 8$ symbols. LoRaWAN wraps the application payload of B bytes in 13 bytes of its own, so $PL = 13 + B$.

Worked through for an SOS

$B = 15$, so $PL = 28$. The bracket is $(224 - 28 + 44)/28 = 8.57$, rounded up to 9; times 5 is 45; plus 8 is 53 payload symbols. With the preamble, $8 + 4.25 + 53 = 65.25$ symbols, and $65.25 \times 1.024 = 66.82$ ms. The same steps give 138.50 ms for 64 bytes and 338.18 ms for 200 bytes. All three match the first draft exactly, and the firmware's own calculation agrees to a microsecond. At SF9 the SOS frame takes 226.3 ms.

5.2 Collisions

Poles do not coordinate. Each transmits when it has something to send, on a channel picked at random. This is the *pure ALOHA* scheme, and its arithmetic is short. Suppose frames of length T start at random at rate λ per second on one channel. A given frame is destroyed if another starts in the T seconds before it or the T seconds during it: a window of $2T$. The number of such starts is Poisson with mean $2\lambda T$, so the frame survives with probability $e^{-2\lambda T}$. Writing $G = \lambda T$ for the offered load,

$$P_{loss} = 1 - e^{-2G}. \tag{5.2}$$

With M channels the load on each is the total airtime per second divided by M.

When frames have different lengths, a frame of length T_i is destroyed by a frame of type j that starts within T_j before it or T_i after it starts. Summing over types,

$$P_{loss,i} = 1 - \exp\Big(-\frac{1}{M}\Big(T_i \sum_j \lambda_j + \sum_j \lambda_j T_j\Big)\Big). \tag{5.3}$$

Long frames are lost more often than short ones. In a 3,000-wearable surge a 25-tag chunk is lost 34% of the time while the short SOS frame is lost 23% of the time.

Two real effects are left out, and both help: a strong frame can survive a weak one (capture), and different spreading factors interfere only partly. The losses in this book are therefore pessimistic.

A note on rules. The first draft spoke of duty-cycle limits. The LoRaWAN regional parameters for IN865 set a maximum payload and transmit power but, unlike the European band, no duty cycle. The binding limits here are the 222-byte payload and the collisions themselves.

CHAPTER 6

The edge filter and the capacity of a cell

A pole hears a wristband a hundred times in a minute. It needs to say so once.

6.1 What a pole does

The edge filter (Algorithm 2 of the white paper)

On each advertisement with signal strength r:
1. Ignore it unless it is 15 bytes with Company_ID `0xFF5A`.
2. If the SOS bit is set, fewer than 10 SOS frames went out this minute, and this identifier was not forwarded in the last 90 s: send the frame now, confirmed, on port 10, with one byte of r appended.
3. If r is below the gate, stop.
4. Update the table entry: smoothed signal $\leftarrow 0.75\cdot\text{old}+0.25\,r$; count the hit; note the time.
5. When the hits reach H (1 on tracks, 3 in hubs) and the arrival is unreported: queue 8 bytes (identifier, counter, signal, flags).

Every 15 s, after a random 200 to 1500 ms: retire entries unheard for 90 s; send queued arrivals in chunks of at most 25 (200 bytes) on port 1; send departures, 4 bytes each, on port 2.
Every 60 s: send a 64-byte Bloom filter of everyone present on port 3.

The table holds 1,024 entries in about 45 kB of static memory. The C implementation and the Python reference are fed the same ten busy minutes, sixty wristbands arriving, leaving and pressing SOS, and must emit identical uplinks.

Step 2's extra byte is new. In the end-to-end simulation of Chapter 9 one SOS was heard by seven poles up to 220 m away, because Coded PHY carries far. Without a signal strength the server would pin the alarm on whichever uplink arrived first.

6.2 A model of one cell

Take N wristbands in the area served by one gateway, with $K = 60$ poles, each wristband reaching a new pole every $\tau = 55$ s on average. Then:

- arrivals at one pole are a Poisson process of rate $N/(\tau K)$, so in a 15 s window a pole collects a Poisson number of arrivals with mean $15N/(\tau K)$: 2.3 for $N = 500$ and 13.6 for $N = 3{,}000$;
- a pole holds, on average, N/K wristbands: 8 to 50.

Four designs turn this into frames.

Raw forwarding. Every ping heard becomes an uplink of about 7 bytes. The first draft's table assumed a ping every 2 s.

Single-event arrivals. One 8-byte uplink per arrival, and every 60 s a list of the 4-byte identifiers present.

15 s batching. Arrivals are collected for 15 s and sent together; the same raw list every 60 s.

Rakshasutra. Arrivals in chunks of at most 25; departure lists; a 64-byte Bloom filter in place of the raw list.

For each, the expected airtime per second is summed over the Poisson distribution of how many items each report carries, using the exact time on air of each frame size, and divided by 8 channels to give G.

Table 6.1 Offered load and frame loss, "G / loss" in percent, for one cell of 60 poles and 8 channels at SF7 [calculated].

Forwarding design	500	1,000	2,000	3,000	Largest frame
Raw forwarding (2 s pings)	176.8 / 97.1	353.6 / 99.9	707.2 / 100.0	1060.8 / 100.0	7 B
Single-event arrivals + raw ID heartbeat	7.6 / 14.1	14.6 / 25.4	28.7 / 43.7	42.9 / 57.6	over 222 B
15 s batching + raw ID heartbeat	4.5 / 8.6	6.7 / 12.5	10.6 / 19.0	14.6 / 25.3	over 222 B
Rakshasutra as specified	7.7 / 14.3	10.2 / 18.4	14.2 / 24.7	18.2 / 30.5	$\leq$ 200 B
Rakshasutra without departure lists	5.1 / 9.7	6.6 / 12.4	9.3 / 17.0	12.0 / 21.3	$\leq$ 200 B

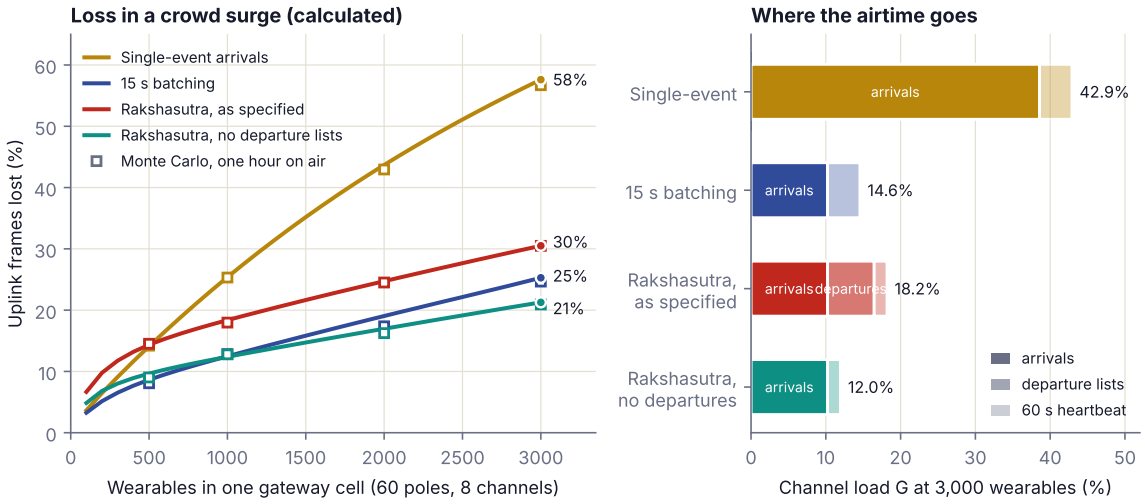


Figure 6.1 Left: loss against the number of wristbands in the cell; squares are the Monte Carlo check. Right: the airtime of each design at 3,000 wristbands, by stream.

6.3 Checking the formula by brute force

A formula for collisions deserves a check that does not use the formula. The simulator gives every pole its real timers, with random phases and jitter, draws its arrivals, puts every frame on a random channel, and marks as lost any two that overlap. Over one simulated hour per case, 695,304 frames in all, the simulated loss stays within 1.7 percentage points of the formula in every case. [simulated]

6.4 What the table says

Batching does most of the work. Sending arrivals together every 15 s instead of one by one cuts loss at 3,000 wristbands from 57.6% to 25.3%. A frame's fixed cost, preamble, header and 13 bytes of LoRaWAN, is paid once for many arrivals.

The Bloom filter is what keeps frames legal. A raw list needs 4 bytes a wristband and overflows 222 bytes beyond 55 wristbands. At $N = 3{,}000$ a pole holds 50 on average, and by

Table 6.2 Frame loss, first draft against this model, for 500 / 1,000 / 2,000 / 3,000 wristbands.

Forwarding design	First draft, loss (%)	This model, loss (%)
Raw forwarding (2 s pings)	97.7 / 99.9 / 100 / 100	97.1 / 99.9 / 100.0 / 100.0
Single-event arrivals + raw ID heartbeat	12.2 / 22.9 / 40.5 / 54.2	14.1 / 25.4 / 43.7 / 57.6
15 s batching + raw ID heartbeat	5.8 / 11.3 / 16.8 / 22.1	8.6 / 12.5 / 19.0 / 25.3
Rakshasutra as specified	4.1 / 8.1 / 12.0 / 15.8	14.3 / 18.4 / 24.7 / 30.5
Rakshasutra without departure lists	–	9.7 / 12.4 / 17.0 / 21.3

the Poisson distribution 22% of poles hold more than 55 at any moment. A crowd sheltering from rain would do far worse. The filter is 64 bytes always. It also costs less airtime than raw lists once a cell holds more than about 960 wristbands; below that, raw lists are shorter.

Departure lists cost more than they save. This was the surprise of the audit. The design as specified loses 30.5% of frames at 3,000 wristbands, *more* than plain batching, because each pole also sends a departure list every 15 s. Yet a departure is already implied: the wristband arrives at the next pole, and it vanishes from the heartbeat. Dropping port 2 cuts the load from 18.2% to 12.0% and the loss to 21.3%.

> **The first draft's headline figure does not stand.** It gave 15.8% loss at 3,000 wristbands and an 82.4% cut in airtime. Counting all the traffic of its own algorithm gives 30.5%, or 21.3% without departure lists, and an airtime cut of 57.6% or 72.1% against single-event forwarding. The design still wins clearly, by a smaller margin than claimed.

The deployed layout is easier than the worst case. Sixty poles to a gateway was a stress test. With 195 LoRa poles on six gateways a cell has about 32 poles, and with 5,000 wristbands spread over the route it holds 637. The loss is then 11.0%, or 7.3% without departure lists; the first draft said "well below 6%".

What a lost frame costs. Reports are unconfirmed. A lost chunk hides up to 25 arrivals at one pole; the wristbands are seen again at the next pole about 55 s later. The chance of missing someone at k poles in a row falls as the k-th power of the chunk loss: with 34% chunk loss, about one in nine is missed twice running.

6.5 The Bloom filter in detail

A Bloom filter is a row of bits, here 512. To add an item, hash it to $k = 3$ positions and set those bits. To ask whether an item is present, check its three bits. If any is clear the item is certainly absent. If all are set it is *probably* present: other items may have set them. After n items a given bit is still clear with probability $e^{-kn/m}$, so

$$P_{false} = \left(1 - e^{-3n/512}\right)^3. \tag{6.1}$$

The three positions come from two hashes, $h_1 + j\,h_2$ for $j = 0, 1, 2$, which is as good as three independent hashes for this purpose. The reference fixes the variants the paper left open: CRC-16/CCITT-FALSE for h_1 and 32-bit FNV-1a for h_2.

The rate is 1.6% at 50 wristbands, 8.7% at 100 and 57% at 300. Sixty-four bytes hold about 41 wristbands at 1% and 78 at 5%. Two consequences. The server must not ask a heartbeat "which of my 10,000 wristbands are here?", since hundreds would falsely match; it asks only about wristbands it already places at that pole. And at the crowded hubs the filter saturates; those poles are on fibre, where a filter ten times larger costs nothing.

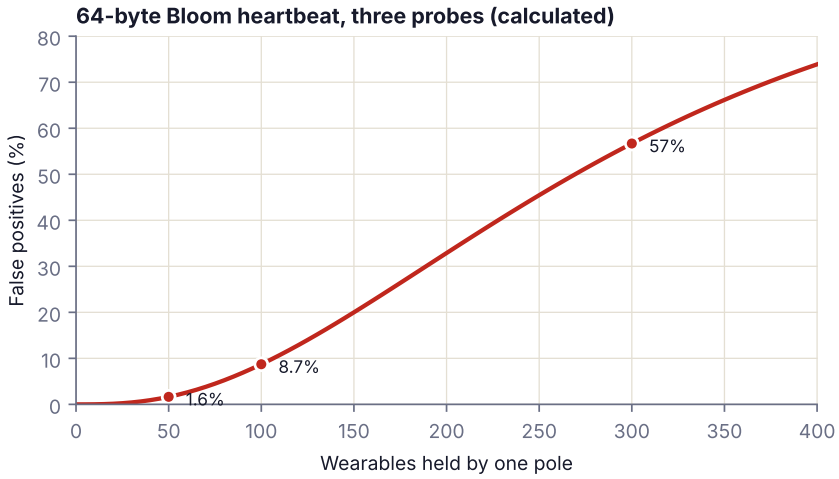


Figure 6.2 False positives of the 64-byte heartbeat [calculated]; a test confirms the curve by direct counting.

6.6 The price of signing sightings

Chapter 3 noted that arrival tuples carry no MIC. Adding it makes a tuple 12 bytes and a full chunk 16 wristbands instead of 25. That is a design choice for the next revision, and the model in the repository can cost it by changing one parameter.

CHAPTER 7

People who move

An alert is only useful if it arrives before the person has walked out of range of it.

7.1 The Class A trap

A LoRaWAN Class A device listens only briefly after each of its own transmissions. To reach a wristband through a Class A pole the server must wait for the pole's next uplink. Add up the worst case: 15 s for the arrival batch, 90 s to be sure the separation is real, up to 60 s for the downlink window, and 8 s for the pole to open a Bluetooth connection to the wristband. That is 173 s. At 1.25 m/s a person walks 216 m in 173 s. By the time the pole tries to connect, the person is two poles further on. The alert fails every time.

7.2 Layer 1: the tether

The first layer needs no network. At pairing, the pad copies the child band's next 96 identifiers, 24 hours' worth and 384 bytes, into the parent tag. The tag listens for them. Four misses in a row, and it sounds its own buzzer.

The scan window

The first draft had the tag open a 5 ms scan window every 5 s. Consider what that window can catch. The band sends an advertisement every 800 ms, plus a random delay of up to 10 ms that Bluetooth requires, at a phase the tag does not know. The packet the tag must catch lasts about 1.2 ms on Coded PHY. A 5 ms window contains a whole packet only if the packet starts in its first 3.8 ms, so the chance is

$$\frac{5 - 1.17}{805} = 0.48\%.$$

Four scans in a row then all miss with probability 98%. The tag would be in alarm almost all the time with the child holding the parent's hand; in simulation, 98% of the time. [simulated]

A window can only be certain to catch a packet if it is as long as the advertising interval, the random delay and one packet together: 811 ms. A receiver drawing 4.6 mA for 811 ms in every 5 s averages 746 μA, twenty-five times the whole budget of the child's band, and would empty the parent tag's cell in about 1.9 months.

> **Design fix.** The tether cannot work as first specified. Two routes are open. The simple one is the long window above, with a rechargeable parent tag. The better one is to make the listening synchronised: Bluetooth 5 periodic advertising lets a receiver wake for a few milliseconds exactly when the next packet is due. That is the route planned for the next revision, and it would also serve the band's own scan for wake tokens.

Where the alarm sounds

The first draft said the tether triggers at about 25 to 30 m. But the tag stops hearing the band where the radio link ends, and by Eq. 4.1 that is beyond 174 m on Coded PHY. To alarm at 30 m the tag must treat *weak* as *gone*: by the same equation, a signal below about -78 dBm. With 4.2 dB of shadowing such a gate will sometimes trip early and sometimes late; requiring four weak readings in a row steadies it.

7.3 Layer 2: a wake token from the poles

The second layer covers what the tether cannot: a parent tag that has failed, a guard starting a search, a separation discovered by the server because two wristbands of one family are at poles far apart. All poles keep their LoRa receivers open continuously (Class C), and the server multicasts an 8-byte token to the poles within two hops of the wristband: five poles on a plain stretch, more at a junction. They broadcast it by Bluetooth, and the wristband buzzes.

The token is the current identifier, a command byte, and the first 24 bits of an AES-CMAC under K_u over the byte `0x03`, the counter and the command. The paper fixed only the sizes; this layout is the reference code's. A wristband accepts a token only for its current slot and at most once in 30 s, so a recorded token is useless after at most 15 minutes and cannot be used to drain a battery quickly.

The first draft promised 2.5 to 5 s from decision to buzzer. That has not been calculated here, and it should not be believed yet, because the wristband hears the token through a 5 ms scan window every 4 s: the same timing problem as the tether, to be solved the same way.

7.4 Passengers at 20 km/h

Electric vehicles carry pilgrims along the Himkoti track at a steady 20 km/h, 5.56 m/s. A passenger crosses a pole's 25 m core zone in 4.5 s.

Many beacons save power with an accelerometer: still, they advertise slowly. But a passenger in a vehicle moving at constant speed feels no acceleration, and the band stays in its slow mode, one advertisement every 2 to 5 s. In 4.5 s it sends two frames, perhaps three. A hub-style rule of three hits then reports at most 25% of passes.

Rakshasutra removes the accelerometer, fixes the interval at 800 ms, and lets poles on transit tracks report after a single decoded frame. A pass then holds five or six frames and needs one. Whether a frame is decoded depends on the crowd: a pole's scanner hears all wristbands in range on the same channel, and two packets that overlap are both lost. With n others in range, each sending a 1.2 ms packet every 800 ms, a frame survives with probability $e^{-2n \times 1.17/800}$. The capture rate is 99.7% with 150 other wristbands in range of the pole, the figure closest to the first draft's 99.4%, and 95.0% with 300. It is not a single number: it falls as the crowd grows. This simple collision model also ignores how an ESP32 scanner follows extended advertisements onto secondary channels, which will cost something more. Scanner capacity in a dense crowd is the second thing to measure.

7.5 An SOS, timed

From button to alarm: wait for the next advertisement, at most 200 ms once SOS is pressed; decode at a pole; one LoRa frame of 66.82 ms; and the backhaul. If the uplink collides, the pole repeats it at SF9 after its acknowledgement window. The wait before a repeat, about 3 s, is assumed here.

In simulation under the 3,000-wristband surge, the first attempt is lost 23% of the time. The median delay is 0.5 s; 92% of alarms arrive within 4 s; 95% within 7.0 s. With 500

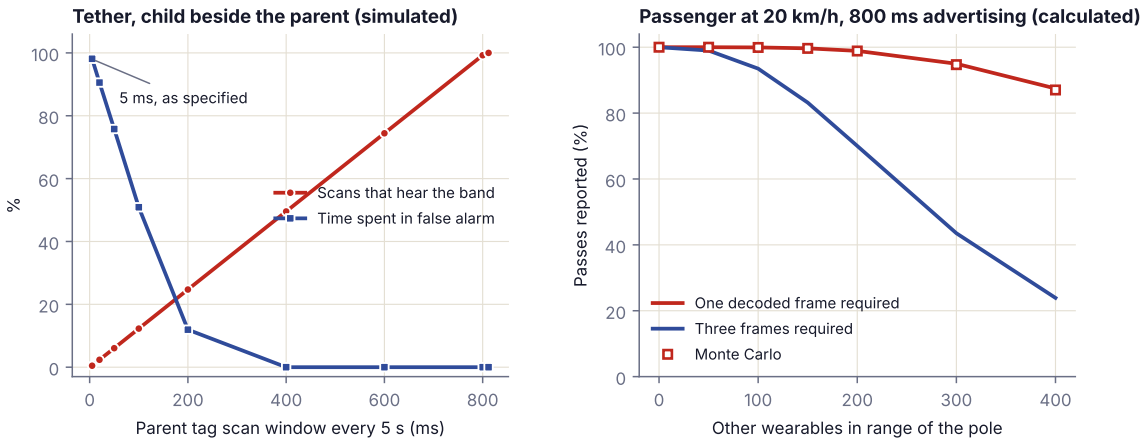


Figure 7.1 Left: the tether, with the child beside the parent [simulated]. The dip to zero false alarms between 400 and 800 ms is a beat between the 5 s and 800 ms periods and should not be relied on. Right: capture of a passenger [calculated, with a Monte Carlo check].

wristbands in the cell 96% arrive within 4 s. On the fibre poles of the hubs, 95% arrive within 0.6 s. [simulated]

The first draft promised “sub-4-second” latency in its abstract and “sub-5-second” in its conclusion. What the model supports is: usually half a second, and within four seconds about nine times in ten under the worst surge. Giving SOS its own channel, kept clear of routine traffic, would close most of the gap and is worth one of the eight.

CHAPTER 8

Power

A coin cell, a year of pilgrimages, and one cold night with the buzzer on.

8.1 The budget

A wristband spends nearly all its time asleep, waking briefly to transmit. Its mean current is the sum, over each activity, of peak current times the fraction of time spent in it. The total,

Table 8.1 Current budget of the wristband, from the first draft's Table V (design values, to be re-derived for the chosen chip).

State	Interval	Peak, mA	On, ms	Mean, μA
Sleep, 32.768 kHz clock and RAM retention	continuous	0.0018	–	1.80
Signed advertisement	every 0.8 s	14.2	1.25	22.19
Scan window for a wake token	every 4 s	4.6	5	5.75
AES-CMAC slot rotation	every 900 s	3.8	0.45	0.00
Total on the trek				29.74

29.74 μA, reproduces the first draft. A CR2450 cell holds 620 mAh. A band spends 14 hours a day on the trek and 10 on the storage rack, where a magnet holds a reed switch open and it draws 0.6 μA. Daily consumption is $14 \times 29.74 + 10 \times 0.6 = 422$ μAh, so the cell lasts

$$\frac{620{,}000}{422} = 1{,}468 \text{ days} = 48.2 \text{ months}.$$

In the cold a lithium cell gives up less of its charge; with the paper's 30% derating at -5 °C the life is 33.8 months. The first draft stated 14.2 and 11.1 months, which do not follow from its own table.

8.2 Coded PHY costs airtime

There is a second problem with the table, and it works the other way. It allows 1.25 ms of transmission per advertisement. Count the bits. A legacy advertisement at 1 Mbit/s with a 15-byte payload is 33 bytes on air, 0.264 ms, sent on three channels: 0.79 ms. That fits 1.25 ms with room for the radio to start up.

But the range in Chapter 4 assumed Coded PHY, which sends every bit eight times. Coded advertising also uses the extended format: a short pointer packet on each of the three primary channels, then the data on a secondary channel. Each packet has a 0.376 ms preamble and access code and then 64 μs per byte:

Packet	PDU bytes	On air (ms)
`ADV_EXT_IND`, three times	9	3×1.168
`AUX_ADV_IND`, once	29	2.448
Total		5.95

At 5.95 ms per event the advertising current rises from 22 to 106 μA, the total to 113.2 μA, and the battery life falls to 12.8 months at 25 °C and 9.0 months at -5 °C. [calculated]

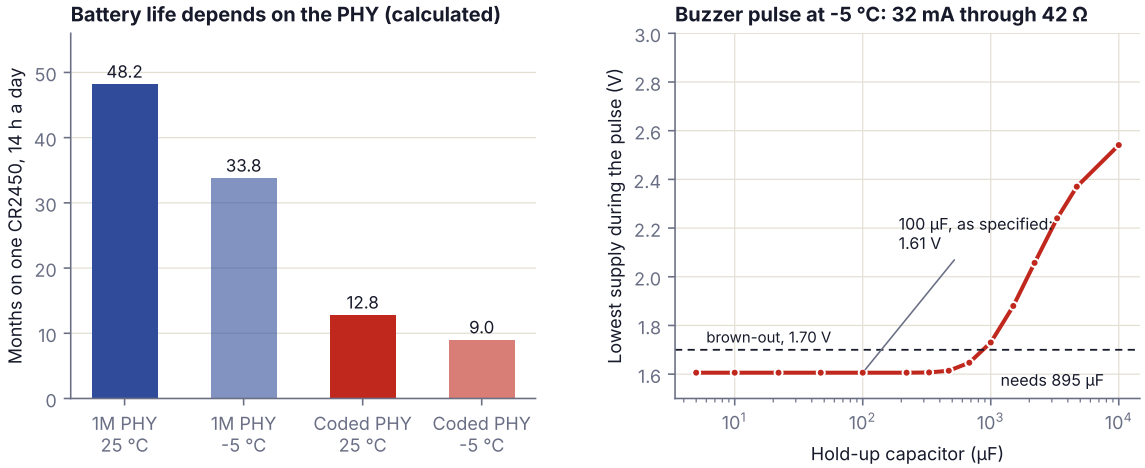


Figure 8.1 Left: battery life. Right: the supply during a buzzer pulse in the cold, against the hold-up capacitor.

By coincidence the Coded PHY figure lands near the first draft's claim. The honest statement is: about a year warm and nine months cold, with range; or about four years warm, without it. A band could use 1M PHY in the dense hubs, where poles are 25 m apart, and Coded PHY on the long tracks.

8.3 The buzzer in the cold

At -5 °C the cell's internal resistance rises to about 42 Ω. The buzzer and a bright LED together draw 32 mA. Ohm's law gives the drop inside the cell: $0.032 \times 42 = 1.34$ V. A cold cell at 2.95 V then delivers 1.61 V, and the radio chip resets below 1.70 V. This much the first draft had right.

Its remedy was a 100 μF capacitor across the supply and a 20% duty cycle, 100 ms on and 400 ms off, with the argument that the *average* current is then 6.4 mA and the drop only 0.27 V. The argument fails because a capacitor averages only over times shorter than its time constant. Model the cell as a voltage source V_0 behind a resistance R, with the capacitor C across the load. During a pulse of current I the supply falls towards $V_0 - IR$ as

$$V(t) = V_0 - IR\bigl(1 - e^{-t/RC}\bigr). \tag{8.1}$$

With $R = 42$ Ω and $C = 100$ μF the time constant is 4.2 ms. A 100 ms pulse is 24 time constants long. The capacitor is exhausted in the first few milliseconds and the supply sits at 1.61 V for the rest of the pulse: a brown-out, exactly as without it. A test in the repository integrates the circuit step by step and agrees with the closed form.

For the supply to stay above 1.70 V through a 100 ms pulse the time constant must be about 38 ms, which needs at least 895 μF, a large part for a wristband. The simpler fix is to draw less: any pulse below 29.8 mA is safe with no capacitor at all. A piezo buzzer and a more modest LED fit under that.

8.4 Which chip

The first draft specified the nRF52811 transmitting at $+8$ dBm. That chip's specified maximum is $+4$ dBm. Parts in the same family that reach $+8$ dBm, such as the nRF52820 and nRF52833, draw about the 14 mA the table assumes. This must be confirmed against the datasheets, and it matters twice: 4 dB less power shortens the Coded PHY range from 174 m to about 113 m, and a different chip changes every current in the table. It is listed as open in the audit.

CHAPTER 9

The route as a graph, and the control room

Pole 173 and pole 174 are neighbours on paper and three kilometres apart on foot.

9.1 Distance on a branching path

Number the poles along the path and it is tempting to measure separation by subtracting pole numbers. On a path that forks, that is wrong. The route is a graph: poles are vertices, adjacent poles are joined by edges weighted with their distance in metres, and the separation of two people is the shortest path between their poles,

$$d_G(p_a, p_b) = \min_{\pi} \sum_{e \in \pi} w(e), \tag{9.1}$$

found by Dijkstra's algorithm. In the model the last pole of the Himkoti track is number 173 and the first pole of the Sanjichhat track is 174. They are 3.0 km apart. The graph has 255 vertices; all distances from one pole take a fraction of a millisecond.

9.2 One family, end to end

The repository contains a simulation that exercises every part at once. A parent with a tag and a child with a band walk the last stretch of the old track and through the Adhkuwari hub at 1.25 m/s. At the junction the parent takes the Himkoti track and the child the Sanjichhat track. The child presses SOS 150 s later. Every advertisement is a real signed frame. Forty-eight poles run the real edge filter. Uplinks from LoRa poles are dropped at random 12% of the time. The server receives 151 uplinks and knows nothing else. The control room flags the separation 105 s after the two part, when the server places them 243 m apart along the track, and issues wake tokens for both. When the child presses SOS, seven poles hear it; the alarm is raised 0.3 s later and placed, by signal strength, at the right pole. The same log is shipped as `examples/adhkuwari_separation.csv`, and

```
rakshasutra trace examples/adhkuwari_separation.csv \
            -registry examples/registry.json
```

replays it.

The 105 s deserve a comment. The server only alerts at 200 m of separation, and its picture of where people are is as coarse as the pole spacing, 108 m on the tracks. Two people must be two poles apart before the alert can fire. This is what Layer 1 is for: the tether would have noticed within 20 s.

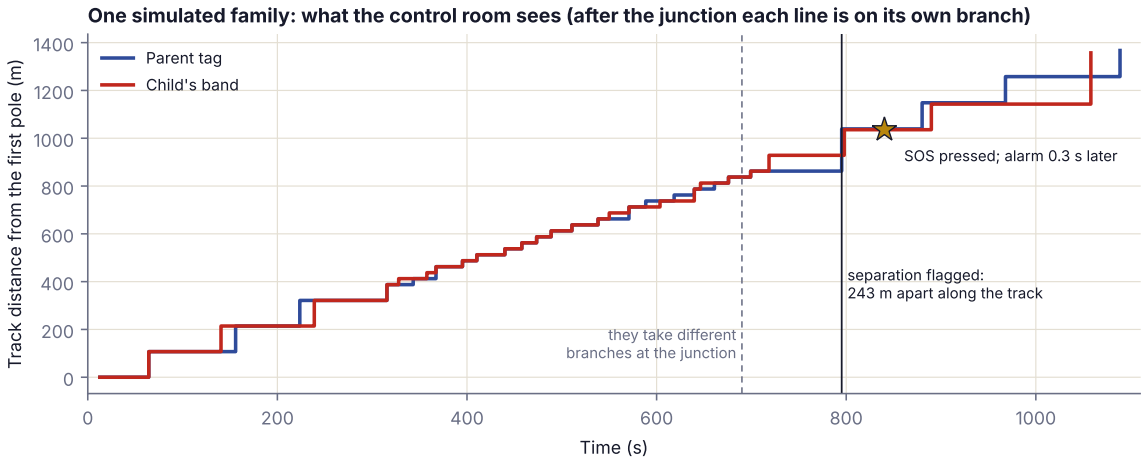


Figure 9.1 What the control room reconstructs [simulated].

9.3 The backend

The white paper specifies an MQTT broker to absorb bursts, in-memory resolution, and batched writes to a time-series database. None of that is built. The reference server is a few hundred lines of Python with a dictionary where the deployed system would have Redis. It exists to pin down behaviour, not performance.

CHAPTER 10

Privacy, law and physical safety

A system that follows children has to be harder on itself than the law requires.

Consent and minimisation. India's Digital Personal Data Protection Act, 2023 requires, in Section 9, verifiable consent from a parent or guardian before a child's personal data are processed, and prohibits tracking or behavioural monitoring of children and advertising targeted at them. Rakshasutra's purpose is the narrow one of safety during a visit. Consent is recorded at registration. The server stores a first name, the band's identifier, a guardian's phone number and the card number. No personal data are broadcast: an advertisement contains a pseudonym, a counter, flags and a battery level. Location trails are deleted within 24 to 72 hours of the band's return. Whether this use falls within an exemption or needs explicit handling under the Act's rules is a legal question to settle with counsel before any pilot; this book does not settle it.

Who can look someone up. Kiosks in the phone-free zones answer only to the family's own card or parent tag, and show only the members bound to it. There is no search by name. A kiosk locks after three failed scans, logs every query, and blanks after 30 s. Control-room staff see more, and their queries must be logged too.

What the design does not protect against. Chapter 3 listed the limits: tracking within one 15-minute slot, and replay of unsigned sightings. A third is the server itself. Whoever holds the master key and the pairing table can follow every wristband. The design assumes an on-premise server under the shrine authority's control, with the key in hardware.

Button cells. A coin cell swallowed by a small child can cause severe internal burns within hours. IEC 60086-4 covers the safety of lithium cells. The band is ultrasonically welded or closed with captive tool-only screws, has no battery door, and is retired, not opened, when its cell is spent. The strap is non-porous and is disinfected between uses.

Radio rules. The 865–867 MHz band is licence-exempt in India for low-power devices within the limits set by the Wireless Planning and Coordination Wing. Any deployment needs type-approved radios within those limits.

CHAPTER 11

The audit

Every checkable number in the first draft, recomputed.

The first draft of White Paper WP-01 was written in October 2026. Its title called the system Rakshasutra and its text called it SurakshaNet; it described its results as "empirical" and spoke of "field propagation measurements". There were none. For the revised paper and this book, each number that can be checked by calculation was recomputed by the code in the repository, using the paper's own equations and parameters.

Verdict	Count	Meaning
reproduced	16	the model gives the first draft's number
differs	16	the model gives another number; the revision uses the model's
design fix	6	the mechanism does not work as specified; the fix is stated
open	4	needs a datasheet check or a measurement

	Claim	First draft	This model	Verdict
A01	Time on air, SOS frame (15 B, SF7)	66.82	66.82 ms	reproduced
A02	Time on air, Bloom heartbeat (64 B, SF7)	138.5	138.5 ms	reproduced
A03	Time on air, 25-tag chunk (200 B, SF7)	338.18	338.18 ms	reproduced
A04	Pseudonyms that fit one raw 222-byte list	55	55 IDs	reproduced
A05	Maximum allowable LoRa path loss, whip above the roof	145.4	145.4 dB	reproduced
A06	BLE Coded PHY margin at 50 m, heavy crowd	28.6	20.3 dB	differs
A07	BLE range at 98% delivery, pole above the roof, 1M PHY, +4 dBm	4.2	4.1 m	reproduced
A08	BLE range at 98% delivery, pole under the roof, 1M PHY	28.5	82.2 m	differs
A09	BLE range at 98% delivery, pole under the roof, Coded PHY	92.0	174.0 m	differs
A10	LoRa range, whip above the roof	5.4-8.8 km	16.1 dB margin at 8 km	open
A11	LoRa range, antenna under the roof	0.85 km	-4.4 dB margin at 8 km	open
A12	Uplink loss, raw forwarding, 500 tags in the cell	97.7	97.1 %	reproduced
A13	Uplink loss, raw forwarding, 3,000 tags in the cell	100.0	100.0 %	reproduced
A14	Uplink loss, single-event arrivals, 500 tags in the cell	12.2	14.1 %	differs
A15	Uplink loss, single-event arrivals, 3,000 tags in the cell	54.2	57.6 %	differs
A16	Uplink loss, 15 s batching, 500 tags in the cell	5.8	8.6 %	differs

	Claim	First draft	This model	Verdict
A17	Uplink loss, 15 s batching, 3,000 tags in the cell	22.1	25.3 %	differs
A18	Uplink loss, Rakshasutra, 500 tags in the cell	4.1	14.3 %	differs
A19	Uplink loss, Rakshasutra, 3,000 tags in the cell	15.8	30.5 %	differs
A20	Uplink loss, Rakshasutra without departure lists, 3,000 tags	15.8	21.3 %	design fix
A21	Reduction in LoRa airtime, 3,000 tags	82.4	57.6 %	differs
A22	Uplink loss in the deployed layout, 5,000 tags on the trek	< 6	11.0 %	differs
A23	Class A worst-case alert delay	173	173.0 s	reproduced
A24	Distance walked during that delay	215	216 m	reproduced
A25	Time an EV passenger spends in a 25 m core zone	4.5	4.5 s	reproduced
A26	EV capture with one decoded frame required	99.4	99.7 %	reproduced
A27	Tether: chance one 5 ms scan hears the child's band	assumed 1	0.48 %	design fix
A28	Tether: separation at which the alert sounds	25-30 m	174 m	design fix
A29	Mean current on the trek (Table V)	29.74	29.74 uA	reproduced
A30	Battery life at 25 °C, from Table V	14.2	48.2 months	differs
A31	Battery life at -5 °C, from Table V	11.1	33.8 months	differs
A32	Transmit time per advertisement on Coded PHY	1.25	5.95 ms	differs
A33	Supply during an unbuffered buzzer pulse at -5 °C	1.61	1.61 V	reproduced
A34	Supply during the pulse with the 100 μF capacitor	2.68	1.61 V	design fix
A35	Chance of forging an SOS in one try	2.33e-10	2.33e-10	reproduced
A36	Chance of forging a wake token in one try	5.96e-08	5.96e-08	reproduced
A37	Pseudonyms are unique in the lookup table	assumed	10% chance of a clash	design fix
A38	Replayed frames are rejected	all	outside the 3-slot window	design fix
A39	Bloom heartbeat false-positive rate	not stated	1.6% at 50 tags, 57% at 300	open
A40	Pole spacing on the LoRa tracks	25-100 m	108 m	differs
A41	Wearable transmit power on the nRF52811	+8 dBm	+4 dBm	open
A42	SOS reaches the control room within 4 s (3,000-tag surge)	all	92% (median 0.5 s)	differs

The notes on each entry are in `AUDIT.md` in the repository, which is generated from the same results file as this table. The command `rakshasutra audit` prints it.

How this happened

The pattern of errors is instructive. The numbers that reproduce are the ones with a formula written beside them: time on air, the path-loss budget, the current sum, the kinematics. The numbers that fail are results stated without a derivation: the capacity table, the ranges, the battery life. And the mechanisms that fail are the ones where an average was trusted in place of a time scale: an average current through a capacitor too small to average it, and a scan window compared with nothing at all. The remedy adopted here is mechanical. No number in this book, the white paper, the README or the website is typed by hand; each is written by script from a results file, and a test fails if a saved result no longer matches the code.

CHAPTER 12

Limits and next steps

What this work is not, and what would make it more.

12.1 Limits

- **Nothing is measured.** Roof loss, the path-loss exponent, body loss, shadowing, the currents: all are design values. Every range and every lifetime depends on them.
- **The route is schematic.** Zone lengths are the white paper's. Pole positions, altitudes within a zone and the split of zone 4 are assumed. It is not a survey.
- **The capacity model is simple.** Poisson movement, pure ALOHA, no capture, one spreading factor. It is pessimistic about collisions and silent about gateway limits and downlink traffic.
- **Bluetooth capacity is barely modelled.** One collision formula on one primary channel. A real scanner following extended advertisements in a crowd of hundreds will do worse.
- **Layer 2 latency is not modelled**, and the wearable's scan for wake tokens has the tether's problem.
- **The pole sketch has not run on hardware**, and there is no wearable firmware. The C core is tested on a host computer only.
- **The cost figure** of INR 225–235 per band is an estimate from the first draft, not a quotation.
- This is research software and a design study. It is not a safety system and must not be relied on as one.

12.2 Next steps

Stop	Output	Status
Reference model	Package, firmware core, simulations, this book, WP-01 revision 2, the audit	Done
Protocol revision	Periodic-advertising tether; signed arrival tuples; no departure lists; SOS channel; masked counter	Next
Bench radio	Roof loss and range under a GI shed; scanner capacity with many beacons; currents of the chosen chip	Planned
Prototype	Ten bands, three poles, one gateway; the pole sketch brought up	Planned
Pilot	One zone, with the shrine authority's permission and a legal review	After that
Peer review	A paper built on measurements	After that

Three rules are fixed in advance. Measured results will be reported beside the calculated ones in this book, whichever way they fall. Calculated and simulated results stay labelled as such. And every number in a report is written from result files by script.

12.3 An invitation

The first measurement needs two Bluetooth development boards, a LoRa pair, and a steel shed. If you have a GI shelter and an afternoon, or experience with dense Bluetooth scanning, or you work on crowd safety at a shrine, the repository’s issue tracker has a form for offering data or field access.

APPENDIX A

Parameters

Every number the models use, with its status. PAPER: a design value from the white paper, not measured. STANDARD: fixed by a specification. DEFINED: an encoding this work had to fix. ASSUMED: no source; chosen so that a calculation can run. There are 97 parameters, of which 10 are assumed and 5 defined here. The same table is `data/params.csv`.

Parameter	Value	Status	Source or note
`trek_km`	22.5 km	paper	WP-01 Sec. III-A, total of the six zones
`poles_total`	255 poles	paper	WP-01 Sec. III-A
`poles_fiber`	60 poles	paper	WP-01 Sec. V-D; zones 3 and 5 on the fibre LAN
`gateways`	6 gateways	paper	WP-01 Sec. III-B
`altitude_min_m`	762.0 m	paper	WP-01 Sec. III-A (2,500 ft)
`altitude_max_m`	2017.0 m	paper	WP-01 Sec. III-A (6,619 ft)
`zone4_himkoti_km`	3.5 km	assumed	split of the 6.5 km of zone 4 between its two parallel branches
`zone4_sanjichhat_km`	3.0 km	assumed	split of the 6.5 km of zone 4 between its two parallel branches
`ble_tx_dbm`	8.0 dBm	paper	WP-01 Eq. 2; note the nRF52811 named in the paper is specified to +4 dBm
`ble_tx_dbm_1m_above`	4.0 dBm	paper	WP-01 Table III, first row
`ble_pl0_db`	40.2 dB	paper	WP-01 Eq. 2, path loss at 1 m
`ble_exponent`	2.15	paper	WP-01 Eq. 2, under the steel roof; not measured
`ble_sigma_db`	4.2 dB	paper	WP-01 Eq. 2, log-normal shadowing
`ble_body_db`	14.0 dB	paper	WP-01 Table III, heavy crowd
`ble_antenna_gain_db`	0.0 dB	assumed	sum of wearable and pole antenna gains; the paper gives none
`ble_sens_1m_dbm`	-96.0 dBm	paper	WP-01 Table III
`ble_sens_coded_dbm`	-103.0 dBm	paper	WP-01 Eq. 2, Coded PHY S=8 on ESP32-S3
`gi_loss_ble_db`	24.1 dB	paper	WP-01 Sec. III-B, roof penetration at 2.44 GHz; not measured
`gi_loss_lora_db`	18.4 dB	paper	WP-01 Sec. III-B, roof penetration at 866 MHz; not measured
`pdr_target`	0.98	paper	WP-01 Table III, 'reliable range' means at least 98% delivery
`lora_tx_dbm`	14.0 dBm	paper	WP-01 Sec. III-B
`lora_coax_db`	0.9 dB	paper	WP-01 Sec. III-B, 1 m of RG-316
`lora_gw_gain_dbi`	5.8 dBi	paper	WP-01 Sec. III-B
`lora_sens_sf7_dbm`	-126.5 dBm	paper	WP-01 Sec. III-B, SX1302
`lora_sens_sf9_dbm`	-131.5 dBm	paper	WP-01 Sec. III-B, SX1302
`lora_freq_mhz`	866.0 MHz	standard	centre of IN865, LoRa Alliance RP002-1.0.4
`lora_body_internal_db`	3.0 dB	paper	WP-01 Table III, internal antenna under the roof
`ridge_loss_db`	20.0 dB	assumed	one knife-edge ridge between pole and gateway
`lora_channels`	8 channels	paper	WP-01 Sec. V-A
`lora_mac_overhead`	13 bytes	standard	LoRaWAN MHDR, FHDR, FPort and MIC

Parameter	Value	Status	Source or note
lora_max_app_bytes	222 bytes	standard	IN865 DR5, repeater-compatible; RP002-1.0.4
cell_poles	60 poles	paper	WP-01 Sec. VI-B
pass_seconds	55.0 s	paper	WP-01 Sec. VI-B, mean time between poles
raw_ping_seconds	2.0 s	paper	WP-01 Table IV, row 1
raw_ping_bytes	7 bytes	assumed	6-byte MAC and 1-byte RSSI per forwarded ping
arrival_tuple_bytes	8 bytes	paper	WP-01 Algorithm 2, line 12
departure_bytes	4 bytes	paper	WP-01 Algorithm 2, line 17
chunk_tags	25 tags	paper	WP-01 Algorithm 2, line 16
departure_ids_per_frame	50 ids	paper	WP-01 Algorithm 2, line 17
flush_seconds	15.0 s	paper	WP-01 Algorithm 2
heartbeat_seconds	60.0 s	paper	WP-01 Algorithm 2
depart_timeout_s	90.0 s	paper	WP-01 Algorithm 2, line 15
rssi_gate_track_dbm	-75.0 dBm	paper	WP-01 Algorithm 2, line 8
rssi_gate_hub_dbm	-65.0 dBm	paper	WP-01 Algorithm 2, line 8
sos_per_minute	10 frames	paper	WP-01 Algorithm 2, line 3
adv_interval_s	0.8 s	paper	WP-01 Sec. IV-C
adv_interval_sos_s	0.2 s	paper	WP-01 Sec. IV-C
slot_seconds	900 s	paper	WP-01 Sec. IV-C
walk_mps	1.25 m/s	paper	WP-01 Sec. V-C
ev_mps	5.56 m/s	paper	WP-01 Sec. V-C (20 km/h)
core_zone_m	25.0 m	paper	WP-01 Sec. V-C, length of track a pole reports as 'here'
class_a_batch_s	15.0 s	paper	WP-01 Sec. V-C
class_a_confirm_s	90.0 s	paper	WP-01 Sec. V-C
class_a_downlink_s	60.0 s	paper	WP-01 Sec. V-C
class_a_gatt_s	8.0 s	paper	WP-01 Sec. V-C
tether_scan_period_s	5.0 s	paper	WP-01 Sec. V-C, Layer 1
tether_scan_window_s	0.005 s	paper	WP-01 Sec. V-C, Layer 1
tether_misses	4 scans	paper	WP-01 Sec. V-C, Layer 1
adv_delay_max_s	0.01 s	standard	Bluetooth Core 5.1, random advDelay of 0 to 10 ms
separation_alert_m	200.0 m	paper	WP-01 Sec. V-C, Layer 2
suspect_speed_mps	8.0 m/s	assumed	above the 5.56 m/s of an electric vehicle; faster sightings are flagged
suspect_min_m	300.0 m	assumed	shorter hops are within radio range of two poles and are not flagged
class_c_downlink_s	1.0 s	assumed	server decision, network server and one Class C downlink
lorawan_retry_s	3.0 s	assumed	wait before a confirmed uplink is repeated (RX2 window plus back-off)
backhaul_s	0.3 s	assumed	gateway to control-room alarm
battery_mah	620.0 mAh	paper	WP-01 Table V, CR2450
battery_cold_derate	0.3	paper	WP-01 Sec. VI-C, capacity lost at -5 °C
battery_r_warm_ohm	15.0 ohm	paper	WP-01 Sec. VI-C
battery_r_cold_ohm	42.0 ohm	paper	WP-01 Sec. VI-C
battery_v_cold	2.95 V	paper	WP-01 Sec. VI-C, open-circuit voltage at -5 °C
brownout_v	1.7 V	paper	WP-01 Sec. VI-C, nRF52811 minimum supply
i_sleep_ua	1.8 uA	paper	WP-01 Table V
i_tx_ma	14.2 mA	paper	WP-01 Table V
t_tx_ms	1.25 ms	paper	WP-01 Table V; matches a 1M PHY event on three channels, not Coded PHY
i_rx_ma	4.6 mA	paper	WP-01 Table V
t_scan_ms	5.0 ms	paper	WP-01 Table V
scan_period_ms	4000.0 ms	paper	WP-01 Table V
i_cmac_ma	3.8 mA	paper	WP-01 Table V
t_cmac_ms	0.45 ms	paper	WP-01 Table V
i_rack_ua	0.6 uA	paper	WP-01 Table V

Parameter	Value	Status	Source or note
active_hours	14.0 h/day	paper	WP-01 Sec. VI-C
i_pulse_ma	32.0 mA	paper	WP-01 Sec. VI-C, buzzer and LED
holdup_uf	100.0 uF	paper	WP-01 Sec. VI-C
pulse_on_ms	100.0 ms	paper	WP-01 Sec. VI-C
pulse_off_ms	400.0 ms	paper	WP-01 Sec. VI-C
company_id	65370	paper	WP-01 Table II, laboratory identifier
bloom_bits	512 bits	paper	WP-01 Algorithm 2
bloom_probes	3	paper	WP-01 Algorithm 2, lines 20-22
active_table	1024 entries	paper	WP-01 Algorithm 2
window_slots	3 slots	paper	WP-01 Sec. IV-D
arbitration_s	15.0 s	paper	WP-01 Algorithm 1, line 12
crystal_ppm	20.0 ppm	paper	WP-01 Sec. IV-D
truncation	leading bytes	defined	Trunc keeps the first bytes of the CMAC, as in RFC 4493
counter_encoding	little-endian	defined	the counter enters every CMAC input as it appears on air
wake_token_layout	id4|cmd1|tag3	defined	the paper fixes 8 bytes and 24 bits of tag only
sos_uplink_layout	frame15|rssi1	defined	one RSSI byte is appended so the server can pick the nearest pole
bloom_hashes	CRC-16/CCITT-FALSE, FNV-1a 32	defined	the paper names CRC16 and FNV1a without variants

APPENDIX B

Reproducing everything

```
git clone https://github.com/0xr4j/rakshasutra.git
cd rakshasutra
pip install -r requirements.txt && pip install --no-deps -e .
make test        # the automated tests; compiles the firmware core too
make results     # every study; a few seconds, all seeds fixed
make generated   # numbers, tables, figures, AUDIT.md
make book whitepaper site
```

Path	Content
`src/rakshasutra/`	the reference model: protocol, radio, capacity, power, route, server, simulations
`firmware/`	the C core, a host test, and the pole sketch
`experiments/`	one script per study
`tests/`	the automated tests
`data/`	parameters, zones and poles
`results/`	saved output of every study
`generated/`	numbers, tables and figures written from the results
`examples/`	a family walk and an attack log to replay
`book/`, `whitepaper/`	the sources of this book and of WP-01
`demo/`, `docs/`	the browser explorer and the website

APPENDIX C

What the tests pin down

- **Cryptography.** AES-128 against the FIPS 197 vector; AES-CMAC against the four RFC 4493 vectors and against an independent library; the frame layout byte by byte; rejection of every single-bit change.
- **Radio and airtime.** The three time-on-air values of the paper; the ALOHA formulas; the capacity table's headline numbers; agreement of the formula with Monte Carlo.
- **The edge filter.** Report-once, chunking at 25, the three-hit rule, departure after 90 s, the SOS rate limit, a bounded table.
- **The server.** The expected counter, the three-slot window, stale and unknown identifiers, forged SOS, signal-strength arbitration, identifier clashes, unpairing.
- **Models.** Link budget, battery life, the RC droop against a time-stepped circuit, kinematics, the route graph.
- **Firmware.** The C core against the Python reference: CMAC, 200 random frames with addresses and wake tokens, verification, time on air, and ten busy minutes of the edge filter under three configurations.
- **The browser explorer.** Its JavaScript kernel against the Python reference.
- **Bookkeeping.** The parameter file matches the package; the shipped examples are what the generator writes; the saved results and the audit match a fresh calculation.

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